Bow Valley College

Course Outline

SODV3302

Systems Development and Object-Oriented Design
Fall 2020 - Spring 2021

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SODV3302 Systems Development and Object-Oriented Design

COURSE DESCRIPTION

In this course learners explore advanced object-oriented application development concepts such as design patterns, threads, and event handling. Learners create applications requiring database manipulation, graphical user interfaces (GUI), and data communication.

REQUISITES	Complete or concurrently enroll in: • DATA3201 - Database Management (3) • SODV3301 - Software Development Techniques (3)
EQUIVALENTS	None
CREDITS	3
HOURS	45
ELIGIBLE FOR PLAR	No

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Design computer programs using exception handling.	1, 2, 3, 4, 5, 6, 7
2	Apply graphical user interface (GUI) elements and event handling to build applications that interact with the user.	1, 2, 3, 4, 5, 6, 7
3	Create computer applications using connectivity classes to interact with a database system.	1, 2, 3, 4, 5, 6, 7
4	Describe techniques for establishing connections and performing data stream operations.	1, 2, 3, 4, 5, 6, 7

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Week 1	Module 01: First class - Course Orientation/Introduction • Syllabus, outline, work ethics, assessments, resources, expectations, workload, instructional methods, course dynamics, D2L, tools • Understand the evolution of object oriented language • Basic OOP concepts • Class, Objects, Fields, Methods, Access Modifiers • Q&A
Week 2	Module 01: Pillars Of OOPs • Abstraction • Encapsulation • Inheritance • Polymorphism • Reading Activity
Week 3	Module 02: Unified Modeling Language • UML Intro • UML Notations • Relationships • Class Diagrams • Object Diagrams
Week 4	Module 02: • UML • Component Diagram • Deployment Diagram • Use case Diagram • State Chart • Activity Diagram • Interaction Diagram
Week 5	Module 03: Design Pattern Intro and Design Principles • Introduction to Design Patterns • Design Patterns Model • Design Principles • SOLID Principles
Week 6	Module 03: Design Pattern Intro and Design Principles • Types of Design Patterns Introduction • Creational Patterns • Factory Pattern • Absolute Factory Pattern • Singleton Pattern
Week 7	Module 04: Design Patterns • Prototype Pattern • Builder Pattern
Week 8	Module 05: Design Patterns • Structural Patterns • Adapter Pattern
Week 9	Reading Week
Week 10	Module 05: Design Patterns • Bridge Pattern • Composite Pattern • Decorator Pattern
Week 11	Module 05: Design Patterns • Facade Pattern • Flyweight Pattern • Proxy Pattern
Week 12	Module 06: Design Patterns • Behavioral Patterns • Chain of responsibility pattern • Command Pattern • Interpreter Pattern
Week 13	Module 06: Design Patterns • Iterator Pattern • Mediator Pattern • Memento Pattern • Observer Pattern
Week 14	Module 06: Design Patterns • State Pattern • Strategy Pattern • Template Pattern • Visitor Pattern • MVC Pattern
Week 15	Presentation

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1, 2, 3, 4	Quizzes	20%
1, 2, 3, 4	Assignments	40%
1, 2, 3, 4	Graded Learning Activities	30%
1, 2, 3, 4	Engagement	10%

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A minimum grade of D is required to pass this course. However, a program may require a higher grade in this course to progress in the program or to meet specific program completion requirements.

Please consult with the program area or contact the program chair for further details. A minimum Grade Point Average of 2.0 is required for graduation.

GRADING SCHEME

Grade	Percentage	Grade Point	Description
A+	95-100	4.0	Exceptional: superior knowledge of subject matter
A	90-94	4.0	Excellent: outstanding knowledge of subject matter
A-	85-89	3.67	
B+	80-84	3.33	
B	75-79	3.0	Very Good: knowledge of subject matter generally mastered
B-	70-74	2.67	
C+	67-69	2.33	
C	64-66	2.0	Satisfactory/Acceptable: knowledge of subject matter adequately mastered
C-	60-63	1.67	
D+	57-59	1.33	
D	50-56	1.0	Minimal Pass
F	Less than 50	0.0	Fail: an unsatisfactory performance

REQUIRED LEARNING RESOURCES

Albahari, B. & Albahari, J. (2017). C# 7.0 in a Nutshell: The Definitive reference (7th ed.). O'Reilly Media.

ISBN: 9781491987650

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Honesty (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.